# Synchronization Markers for DNAFormer: Stage 4 Results

FYP, DNA data storage · 25 October 2026

Spacing synchronization markers through the strand raised fine-tuned DNAFormer's exact recovery at 8% deletion from **41.0%** to **52.4%**, against a control with identical payload and marker content.

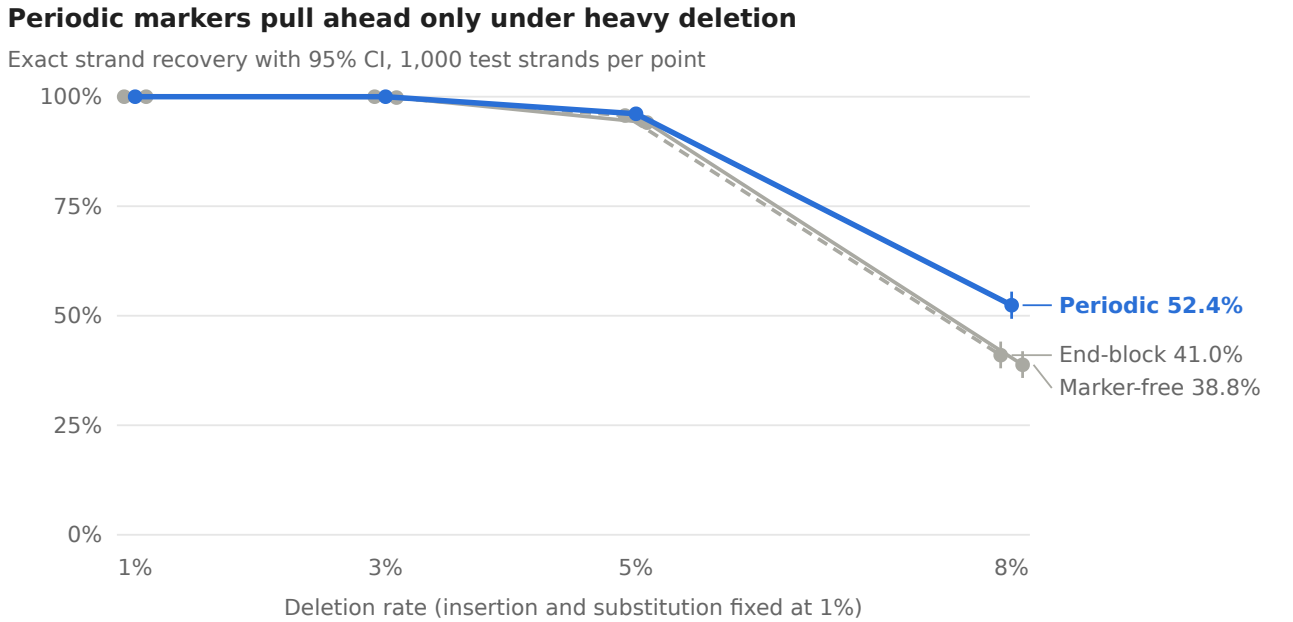
## Question and setup

Do periodic markers (`ACGTCAGT` after every 24 data bases) help a fine-tuned DNAFormer recover 128-base strands when deletions are frequent?

| Version | Layout of the 128 bases | Data bases |
|---|---|---|
| Marker-free | 128 data | 128 |
| Periodic | 4 × (24 data + 8 marker) | 96 |
| End-block (control) | 96 data, then the 32 marker bases together | 96 |

- **Data:** DNAFormer's own simulator; 10 noisy reads per strand; insertions and substitutions fixed at 1%; deletions at 1, 3, 5 and 8%. The versions share the same payloads and noise seeds.
- **Training:** each version fine-tuned from the same pretrained checkpoint with identical settings (10 epochs × 625 steps, best epoch by validation).
- **Scoring:** 1,000 unseen test strands per deletion level; markers removed, only data bases scored.

## Results




| Version | Data bases | 1% | 3% | 5% | 8% | Errors at 8% (NED) | Data recovered at 8% |
|---|---|---|---|---|---|---|---|

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|---|---|---|---|---|---|---|---|
| | | | | | | | **(bases per 1,000 strands)** |
| Marker-free | 128 | 100% | 99.8% | 94.1% | 38.8% | 0.020 | 49,664 |
| **Periodic markers** | 96 | 100% | 100% | 96.1% | **52.4%** | **0.015** | **50,304** |
| End-block control | 96 | 100% | 100% | 95.7% | 41.0% | 0.024 | 39,360 |

Exact strand recovery of fine-tuned DNAFormer by deletion rate; 1,000 unseen test strands per level; markers removed before scoring.

## Findings

1. **Marker spacing drives the gain.** Periodic and end-block carry the same 96 data and 32 marker bases; only placement differs. At 8% deletion periodic recovers 52.4% [49.3, 55.5] vs 41.0% [38.0, 44.1]: the intervals do not overlap.
2. **A shorter payload alone barely helps.** End-block vs marker-free at 8% is 41.0% vs 38.8% (overlapping intervals), and its edit distance is worse (0.024 vs 0.020).
3. **The effect appears only under heavy deletion.** All versions are near-perfect at 1–3%; at 5% they sit within 94–96% with overlapping intervals.
4. **Markers just pay for their 25% overhead.** At 8%, periodic recovers 50,304 data bases per 1,000 strands vs 49,664 marker-free (+1.3%).
5. **Fine-tuning itself helps every version.** At 8%, exact recovery rose from about 8% (pretrained) to 39–52%.

## Is it a worthwhile finding?

Yes, for an FYP: it is a clean, controlled answer with an honest catch.

- **Why it counts:** the end-block control isolates synchronization from the shorter payload, so the claim is causal, not just "markers scored higher". It shows a classic coding idea (watermark-style markers, Davey and MacKay 2001) still helps a strong learned decoder.
- **The catch:** on stored data per strand the markers only break even (+1.3% at 8%), so they are not yet a practical win.
- **Limits:** simulated noise only; one marker sequence, one spacing (24) and one strand length; one training run per version; 10,000 training strands.
- **Before final claims:** Stage 5 adds paired tests on the same strands across versions and the final graphs. Worth testing next: other marker spacings or shorter markers, which could turn break-even into a net gain.